

Mamdani Fuzzy Room Temperature Controller

Temperature + humidity Mamdani FIS · Fuzzy PI · ON/OFF baseline

Generated 08 October 2026

Summary

A Mamdani fuzzy inference system (temperature + humidity -> AC power + fan speed; AND = min, implication = min, aggregation = max, centroid defuzzification) is compared with an improved incremental Fuzzy PI controller and an ON/OFF thermostat ($23.5\text{ }^{\circ}\text{C} \pm 1\text{ }^{\circ}\text{C}$) on a first-order room model.

- Worked example $32\text{ }^{\circ}\text{C}$ / 75 %: only rule 9 fires; ac_power = 76.67 %, fan_speed = 84.44 % (matches the hand calculation and skfuzzy.control).
- The 9-rule Mamdani controller is smooth (1 switching event) but settles outside the comfort zone under strong outdoor loads (proportional-type offset).
- The Fuzzy PI controller removes that offset, keeps 1-5 switching events, and stays in the comfort zone during occupied hours of the 24-hour day.
- The ON/OFF thermostat reaches the zone fastest but switches 10-180 times.
- Occupancy eco setback saves electricity on the 24-hour day; the ESP32 lookup table reproduces the FIS with a mean error of 0.058 % (max 5.02 %).

Main comparison (no noise, no occupancy control)

Scenario	Controller	Settling time	Max overshoot (°C)	SS error (°C)	Energy (kWh)	Cost (₹)	Switching events	Comfort (% of occupied time)	Final T (°C)	Energy saving vs ON/OFF (%)
A) Hot humid day	Fuzzy (Mamdani)	not settled	0.0	1.74	1.65	13.2	1	0.0	25.24	14.0
A) Hot humid day	Fuzzy PI	29 min	0.38	0.38	1.954	15.63	1	75.8	23.13	-3.3
A) Hot humid day	ON/OFF thermostat	25 min	0.23	0.55	1.898	15.18	19	79.2	23.66	-
B) Cold day	Fuzzy (Mamdani)	not settled	0.0	2.81	1.521	12.17	1	0.0	20.69	21.1
B) Cold day	Fuzzy PI	28 min	0.0	0.0	1.943	15.55	1	76.7	23.5	-1.1
B) Cold day	ON/OFF thermostat	24 min	0.22	0.52	1.976	15.81	20	80.0	23.13	-
C) Mild day	Fuzzy (Mamdani)	6 min	0.0	0.24	0.52	4.16	1	95.0	23.82	-8.3
C) Mild day	Fuzzy PI	7 min	0.06	0.06	0.561	4.49	1	94.2	23.45	-17.6
C) Mild day	ON/OFF thermostat	3 min	0.32	0.52	0.468	3.74	10	97.5	24.36	-
D) 24-hour summer day (office)	Fuzzy (Mamdani)	not settled	2.62	1.63	10.241	81.93	1	22.4	21.73	-8.9
D) 24-hour summer day (office)	Fuzzy PI	10 min	0.52	0.28	10.315	82.52	5	100.0	23.18	-11.3
D) 24-hour summer day (office)	ON/OFF thermostat	6 min	0.45	0.42	9.204	73.63	180	100.0	24.38	-

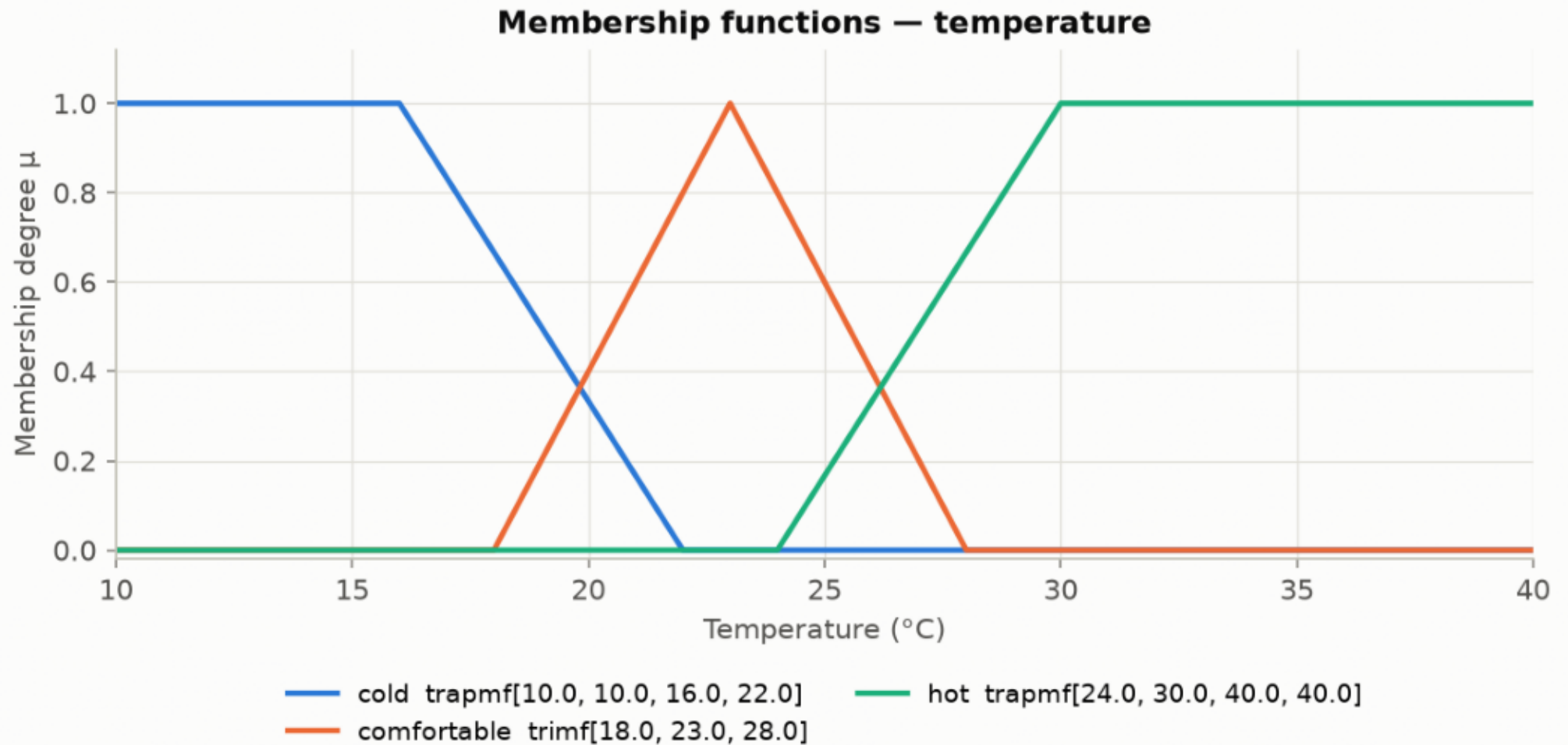
Sensor-noise experiment

Scenario	Controller	Settling time	Max overshoot (°C)	SS error (°C)	Energy (kWh)	Cost (₹)	Switching events	Comfort (% of occupied time)	Final T (°C)	Energy saving vs ON/OFF (%)
A) Hot humid day + noise	Fuzzy (Mamdani)	not settled	0.0	1.78	1.648	13.18	1	0.0	25.26	11.7
A) Hot humid day + noise	Fuzzy PI	29 min	0.56	0.35	1.951	15.61	1	75.8	23.17	-6.1
A) Hot humid day + noise	ON/OFF thermostat	25 min	0.36	0.58	1.846	14.77	18	79.2	24.62	-
B) Cold day + noise	Fuzzy (Mamdani)	not settled	0.0	2.96	1.504	12.03	1	0.0	20.51	23.0
B) Cold day + noise	Fuzzy PI	28 min	0.23	0.08	1.936	15.49	1	76.7	23.4	0.5
B) Cold day + noise	ON/OFF thermostat	24 min	0.38	0.52	2.002	16.02	19	80.0	23.63	-
C) Mild day + noise	Fuzzy (Mamdani)	6 min	0.0	0.4	0.497	3.98	27	95.0	24.03	2.4
C) Mild day + noise	Fuzzy PI	7 min	0.3	0.07	0.572	4.58	19	94.2	23.41	-13.7
C) Mild day + noise	ON/OFF thermostat	3 min	0.42	0.42	0.494	3.95	12	97.5	23.84	-

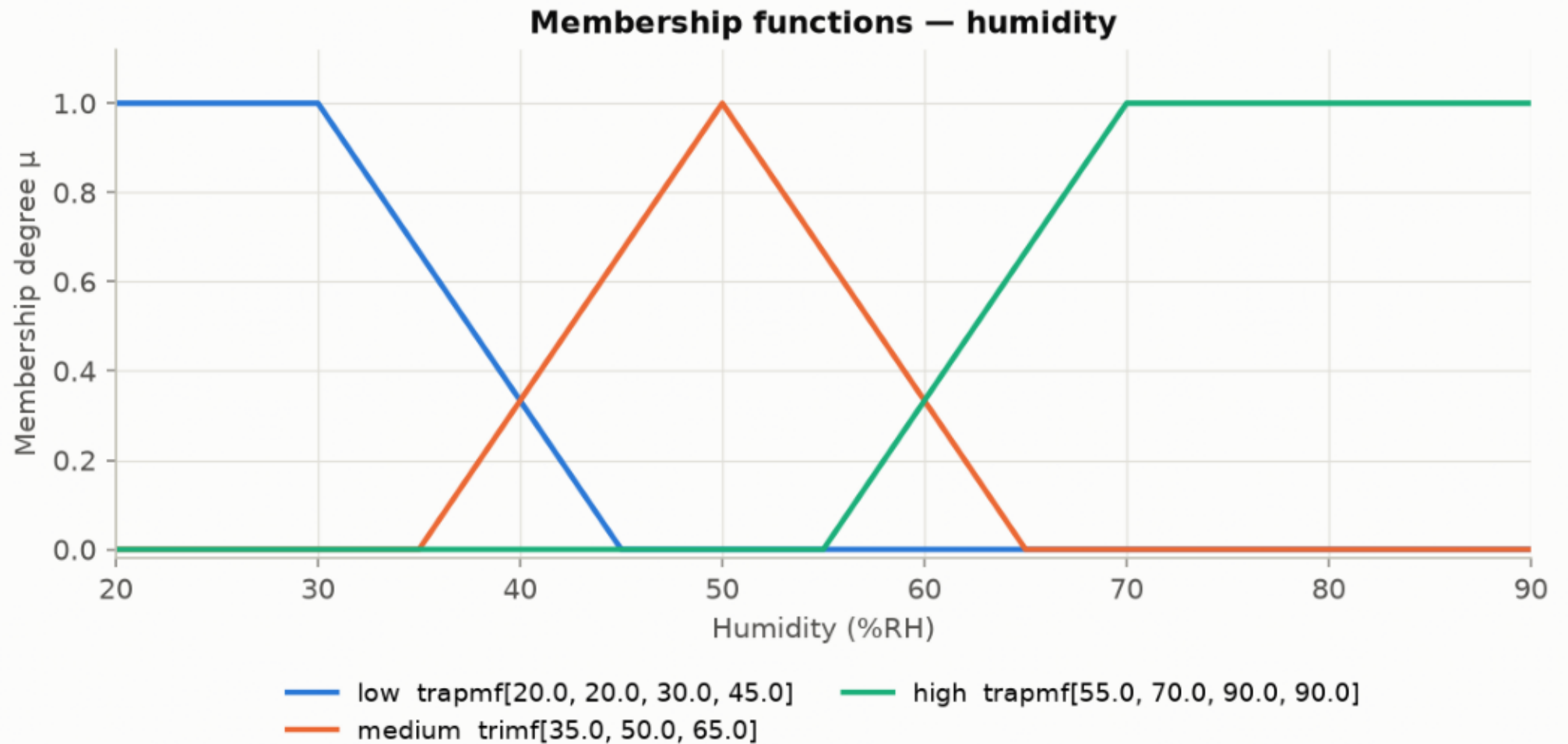
Occupancy experiment (scenario D)

Scenario	Controller	Settling time	Max overshoot (°C)	SS error (°C)	Energy (kWh)	Cost (₹)	Switching events	Comfort (% of occupied time)	Final T (°C)	Energy saving vs ON/OFF (%)
D) always on	Fuzzy (Mamdani)	not settled	2.62	1.63	10.241	81.93	1	22.4	21.73	-8.9
D) always on	Fuzzy PI	10 min	0.52	0.28	10.315	82.52	5	100.0	23.18	-11.3
D) always on	ON/OFF thermostat	6 min	0.45	0.42	9.204	73.63	180	100.0	24.38	-
D) with occupancy eco	Fuzzy (Mamdani)	not settled	0.03	3.26	6.48	51.84	8	23.3	26.54	8.5
D) with occupancy eco	Fuzzy PI	not settled	0.49	3.0	7.537	60.3	6	100.0	26.35	-8.1
D) with occupancy eco	ON/OFF thermostat	not settled	0.31	3.43	7.02	56.16	130	100.0	26.67	-

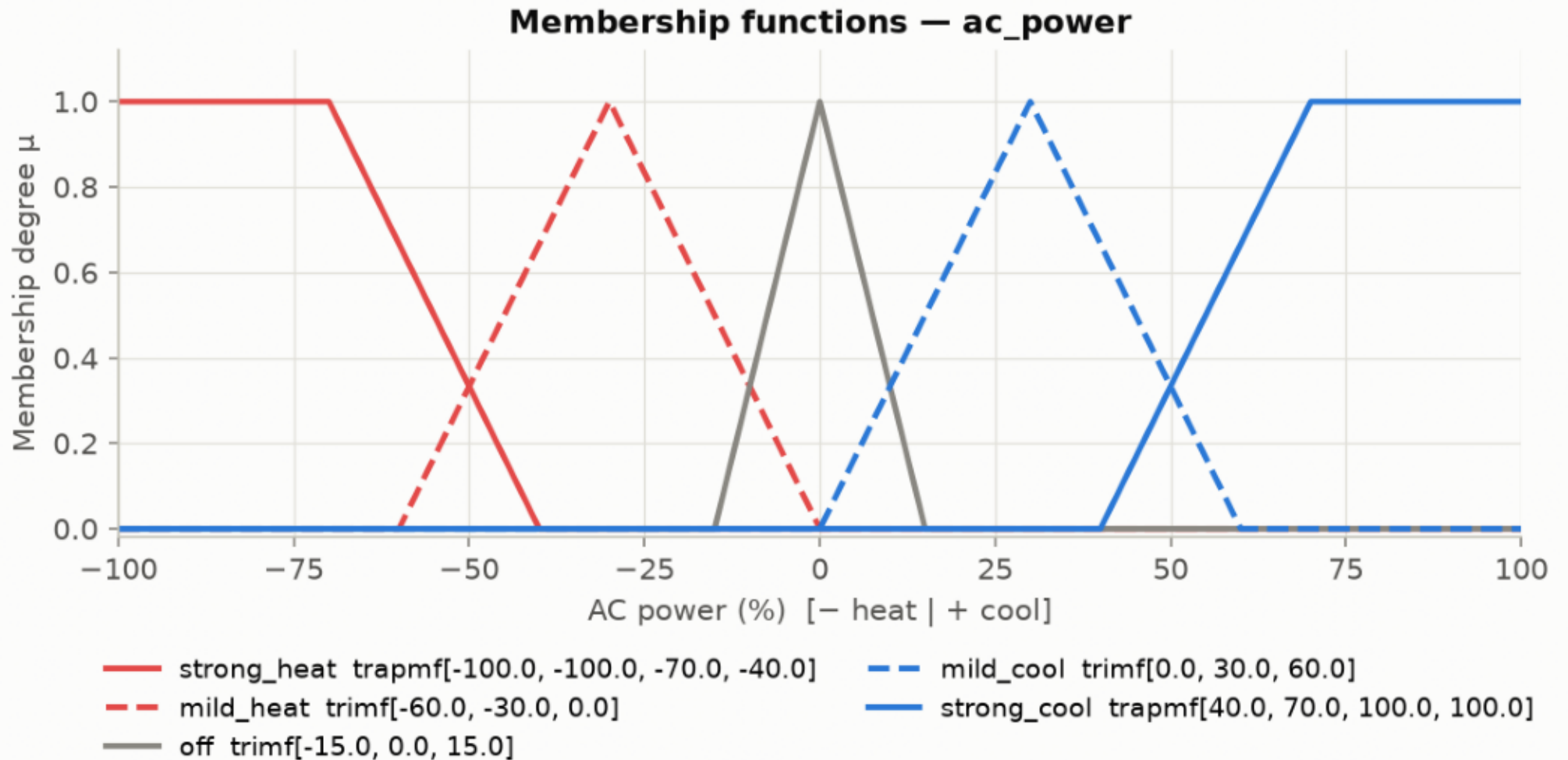
Membership functions — temperature



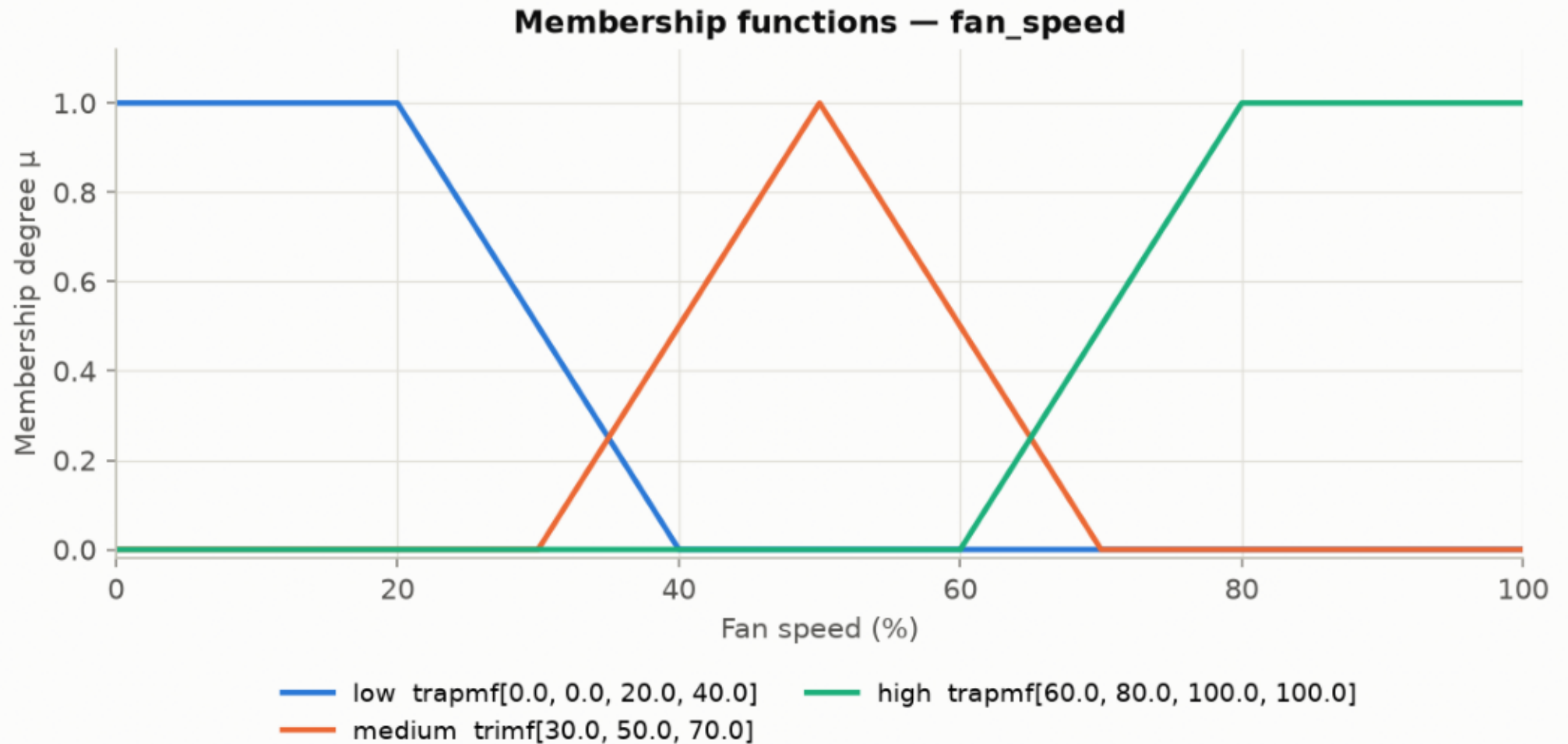
Membership functions — humidity



Membership functions — ac_power

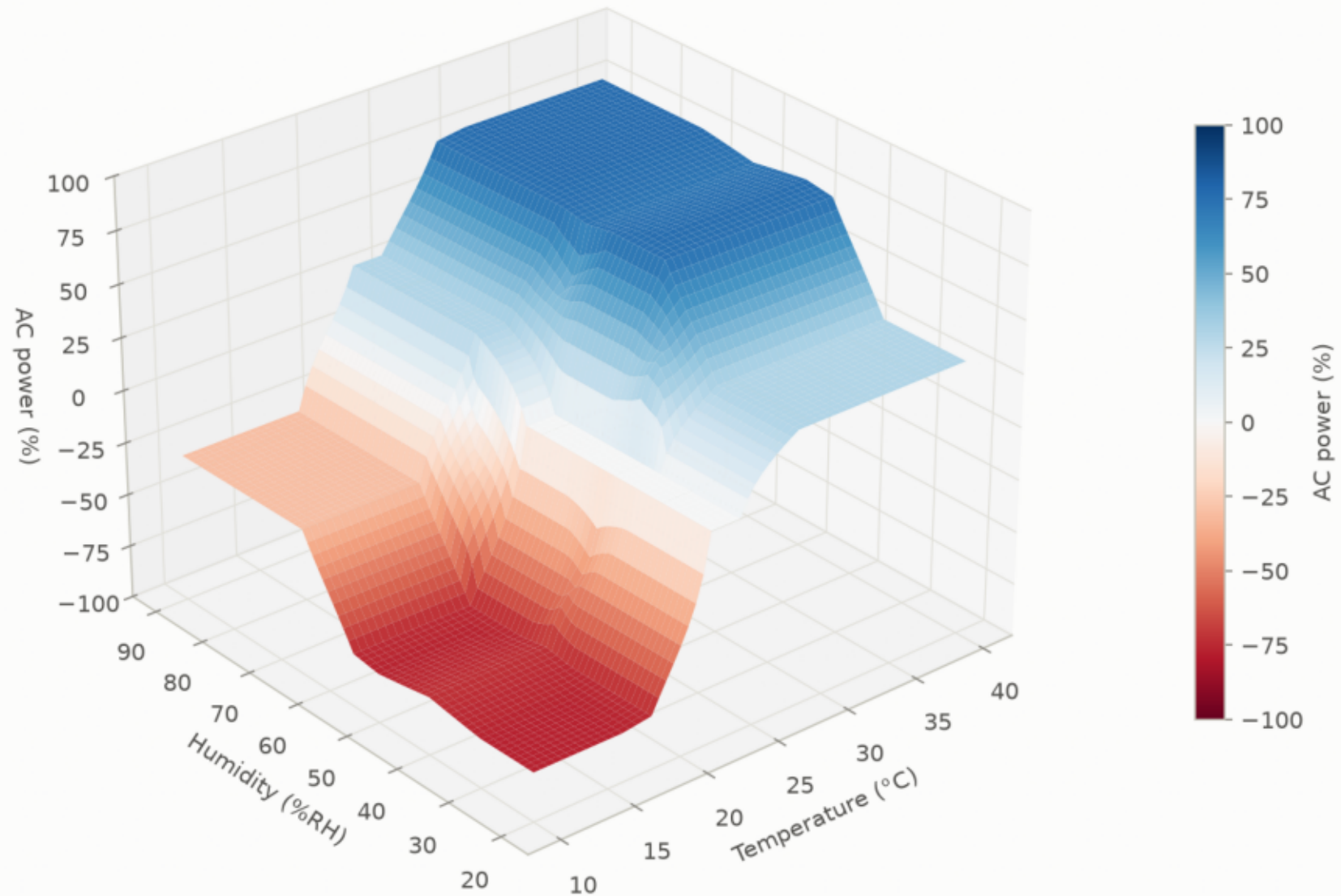


Membership functions — fan_speed



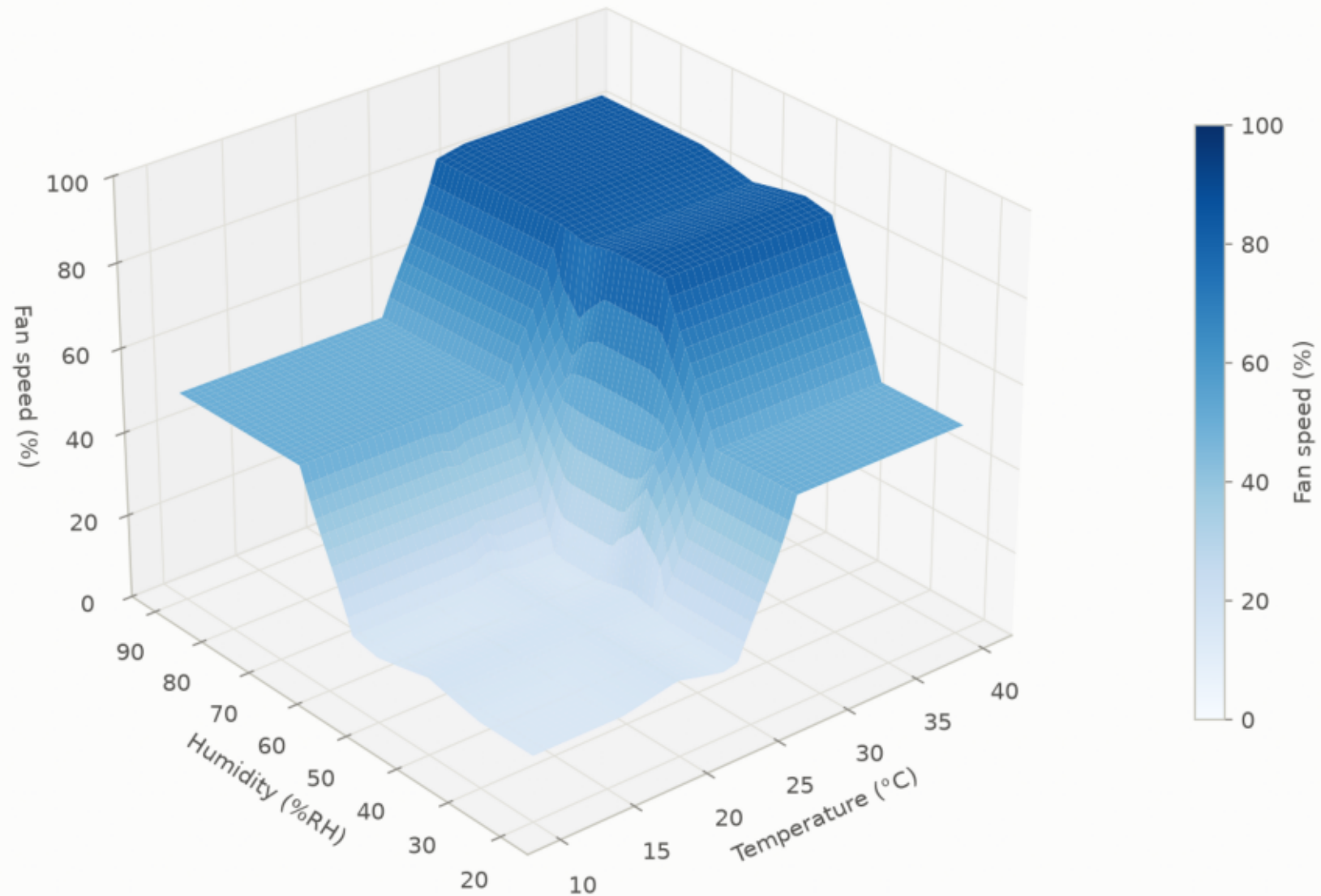
Control surface — ac_power

Control surface — temperature × humidity → ac_power



Control surface — fan_speed

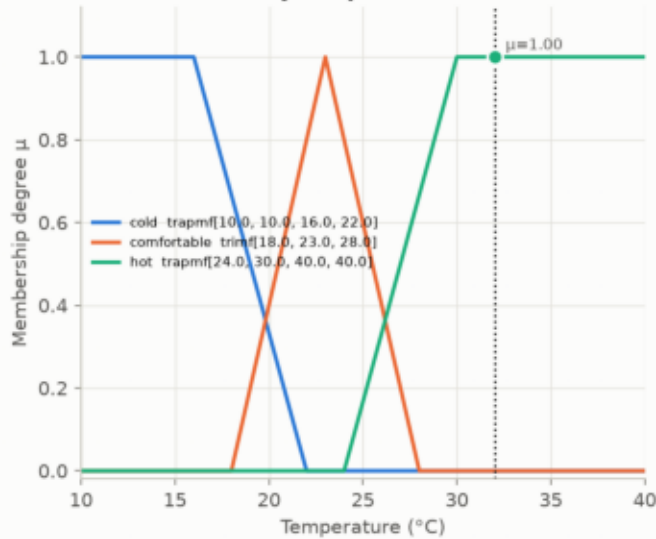
Control surface — temperature × humidity → fan_speed



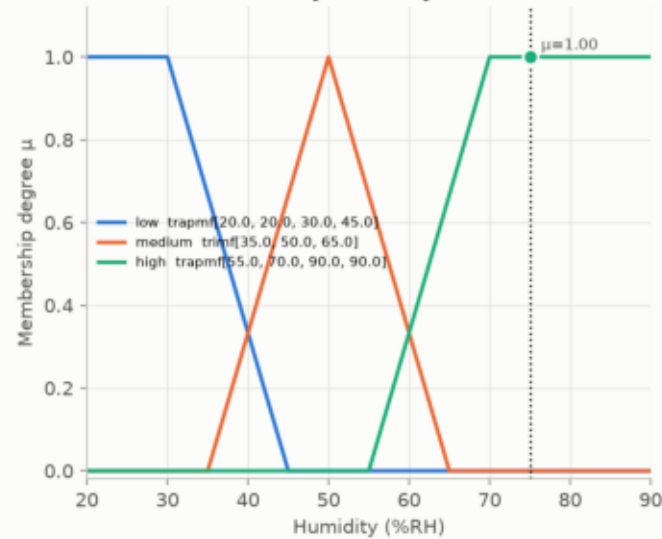
Worked example (32 °C, 75 %)

Worked example — T = 32.0 °C, H = 75.0 % → ac_power = 76.67 %, fan_speed = 84.44 %

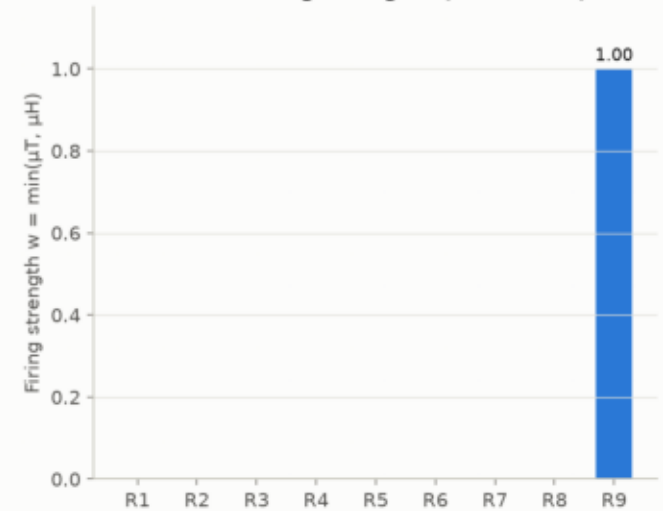
1. Fuzzify temperature = 32 °C



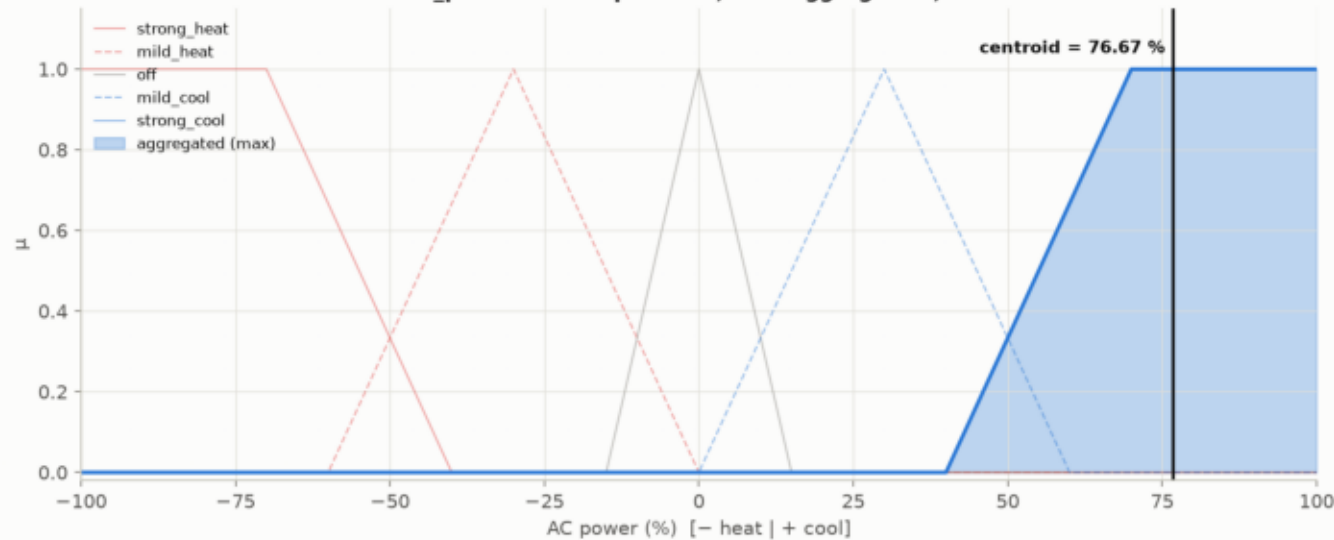
1. Fuzzify humidity = 75 %



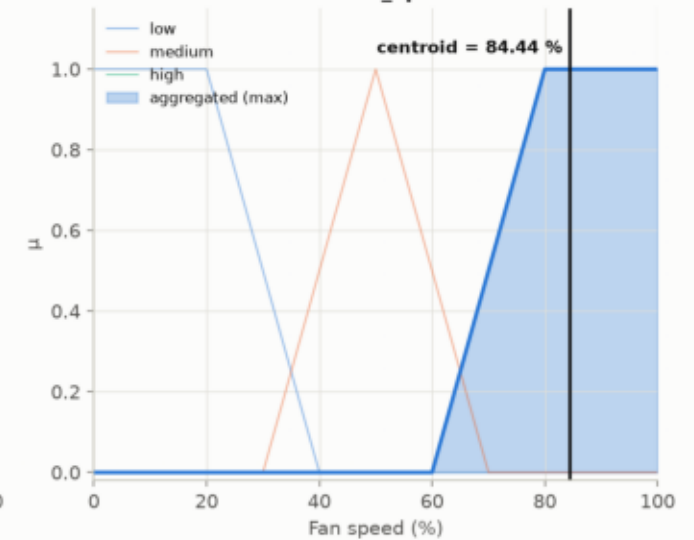
2. Rule firing strengths (AND = min)



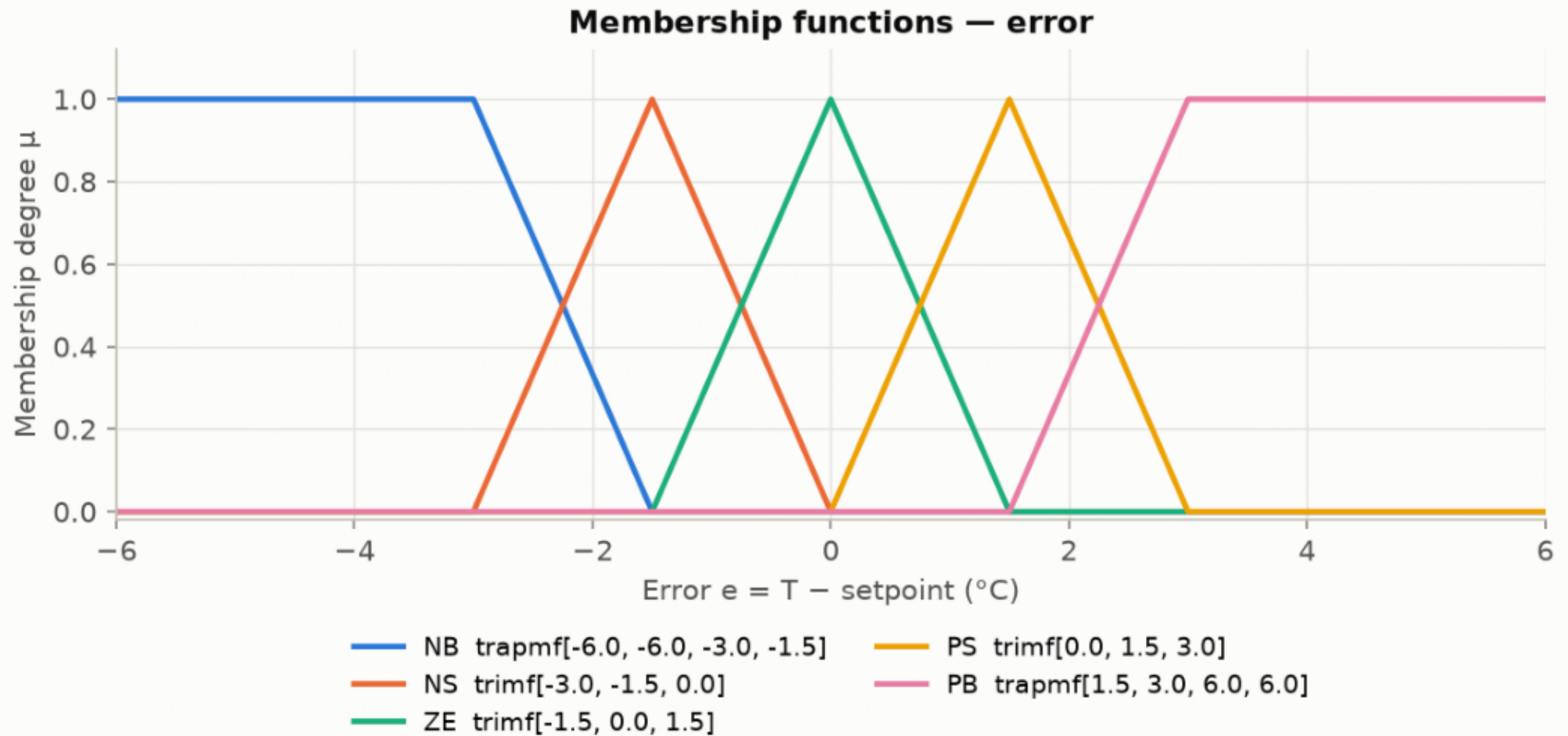
3-4. ac_power: min-implication, max-aggregation, centroid



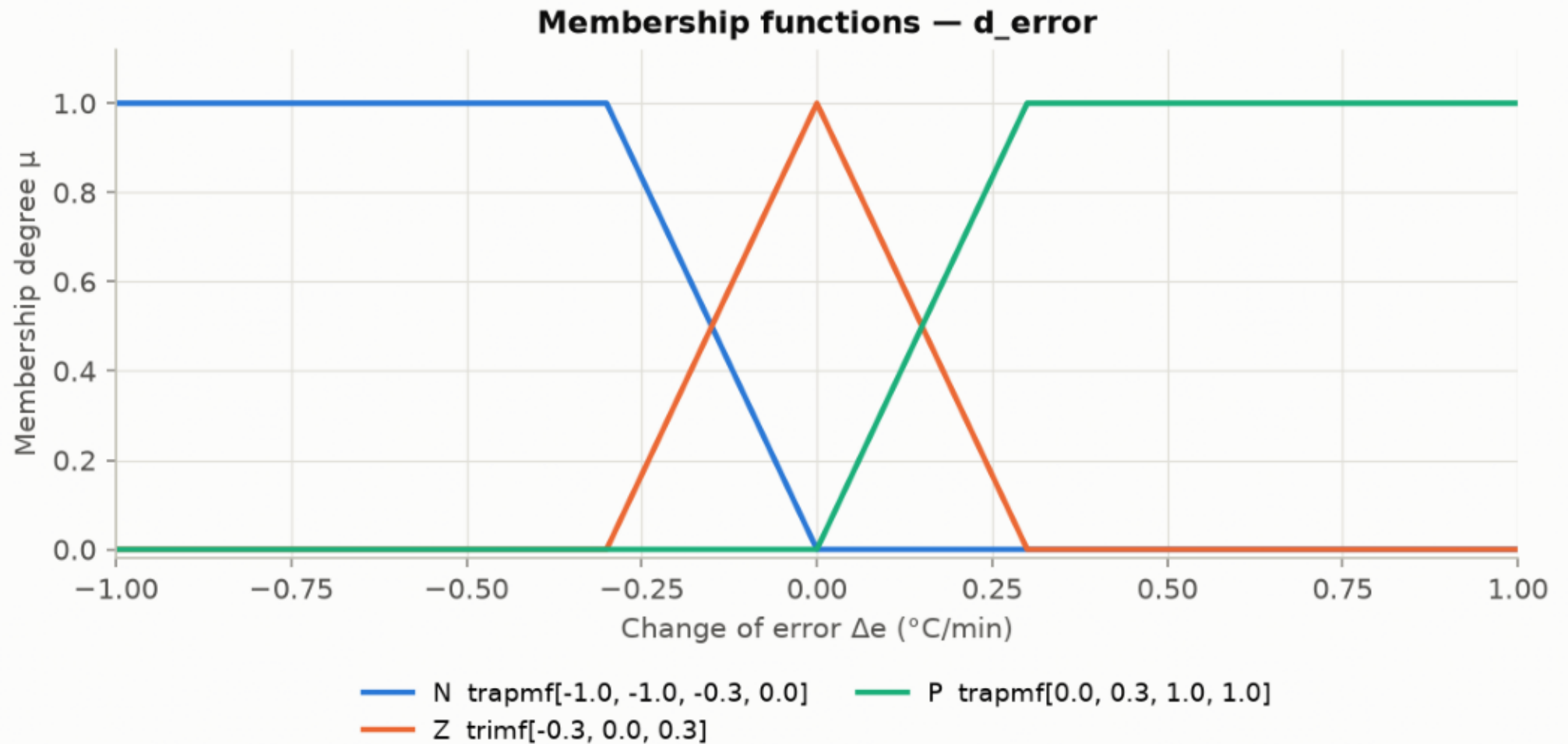
3-4. fan_speed



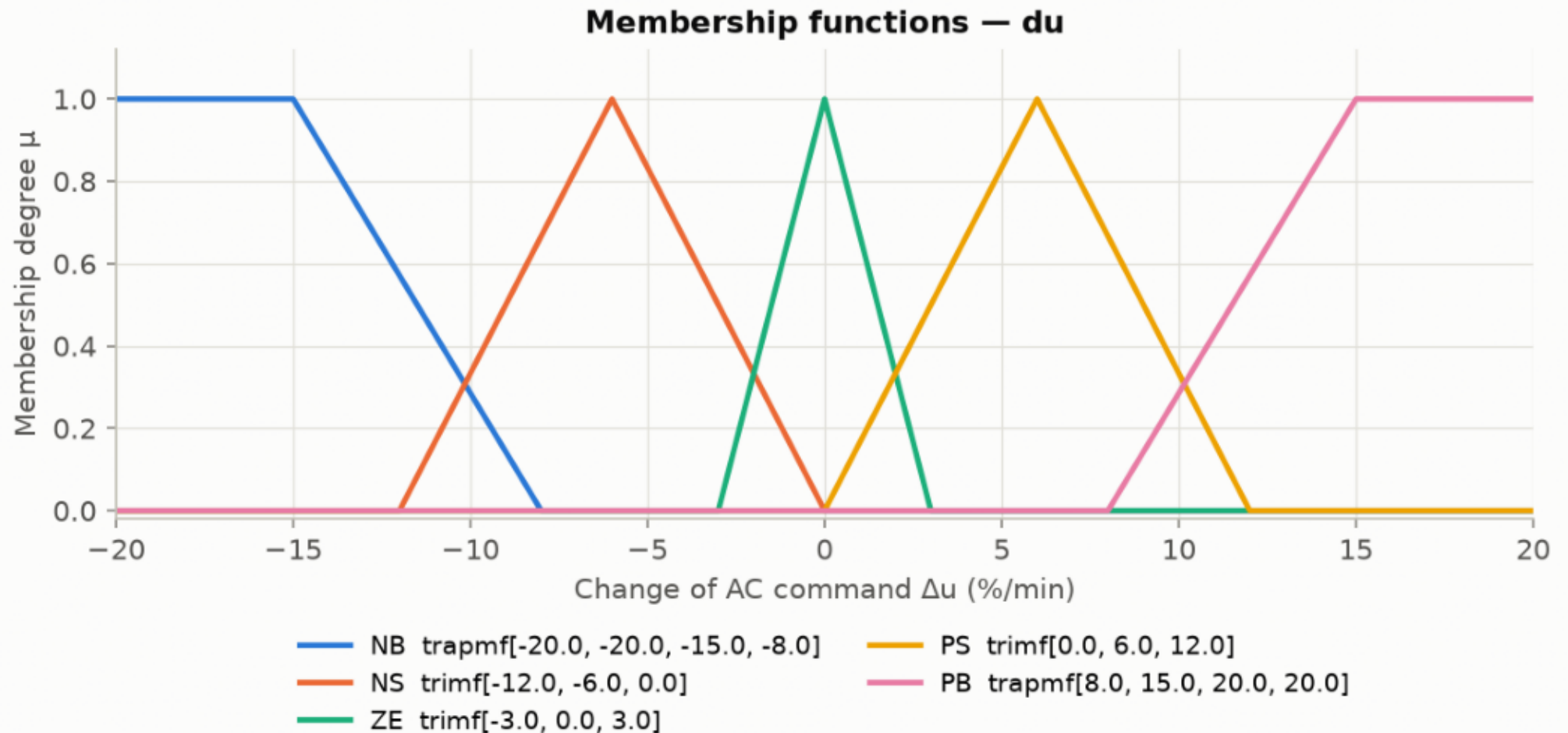
Fuzzy PI membership functions — error



Fuzzy PI membership functions — d_error

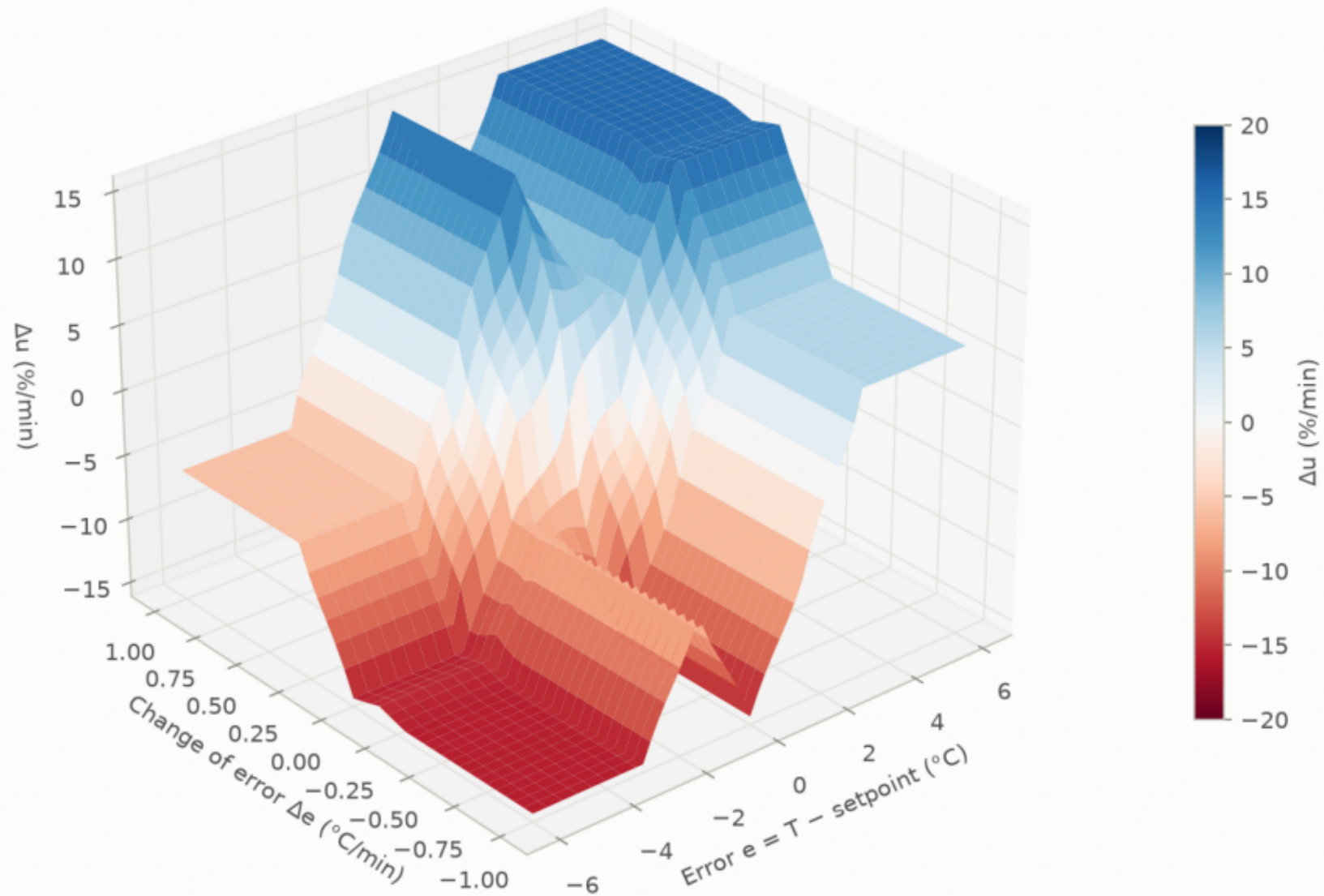


Fuzzy PI membership functions — du

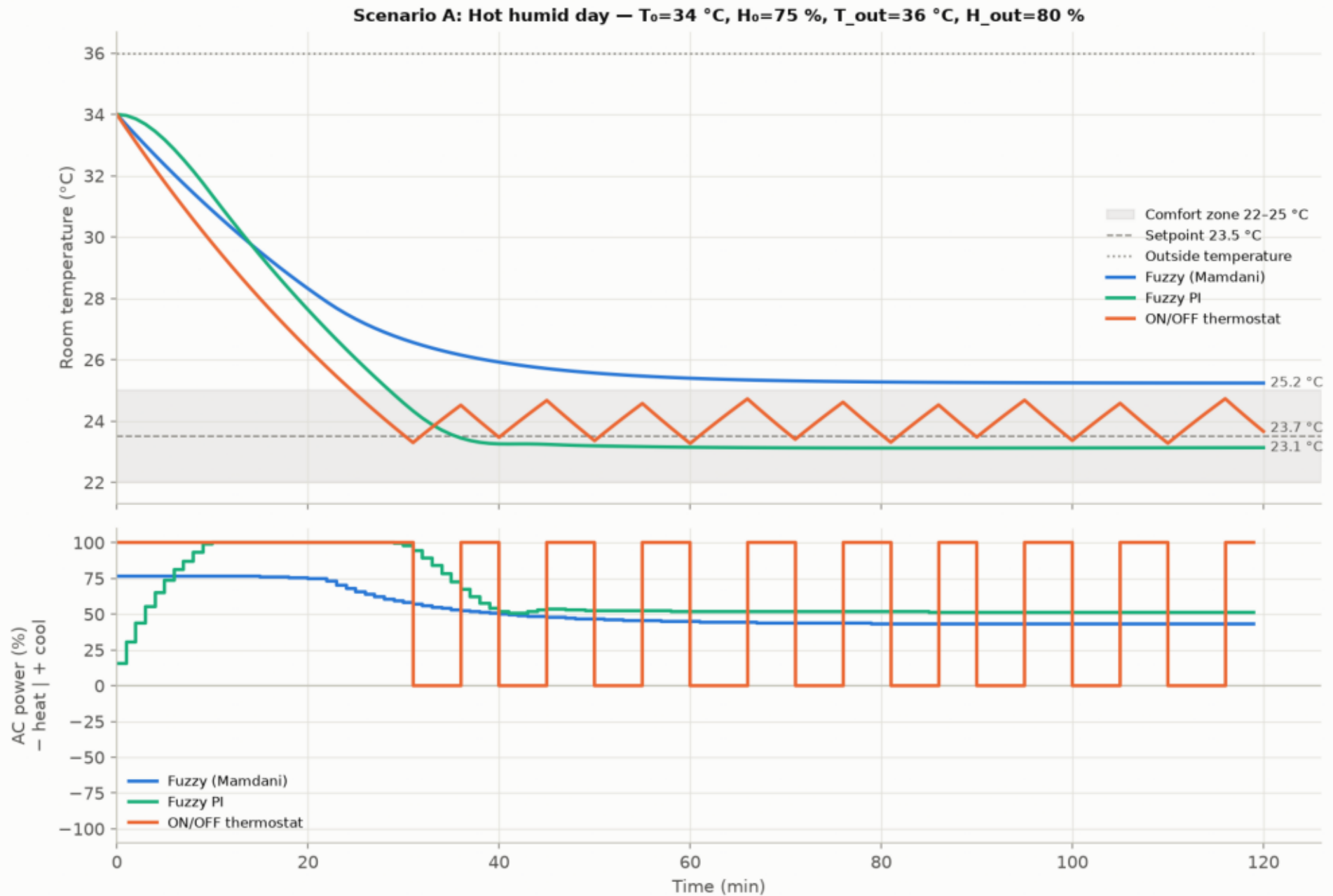


Fuzzy PI control surface

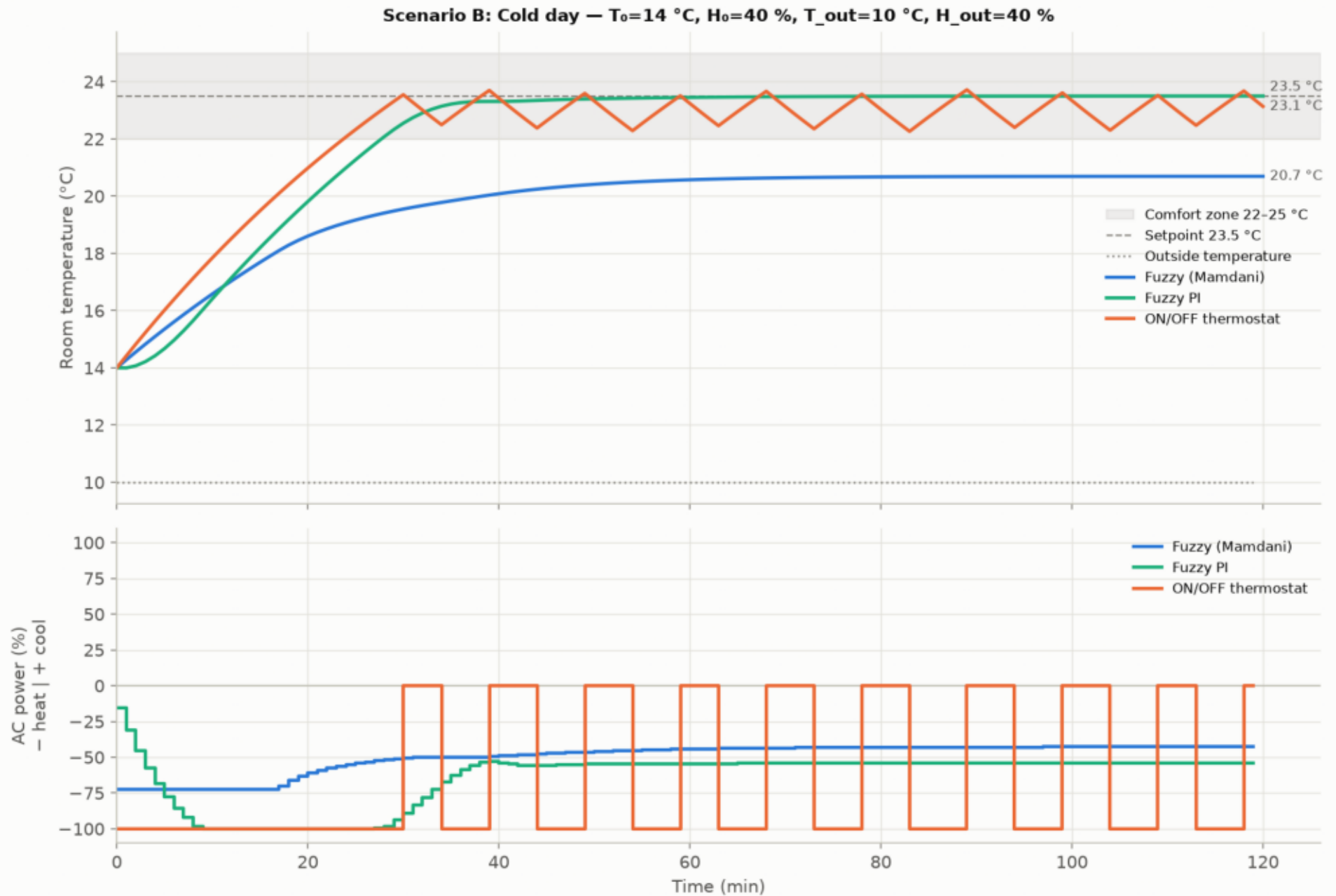
Fuzzy PI control surface — $(e, \Delta e) \rightarrow \Delta u$



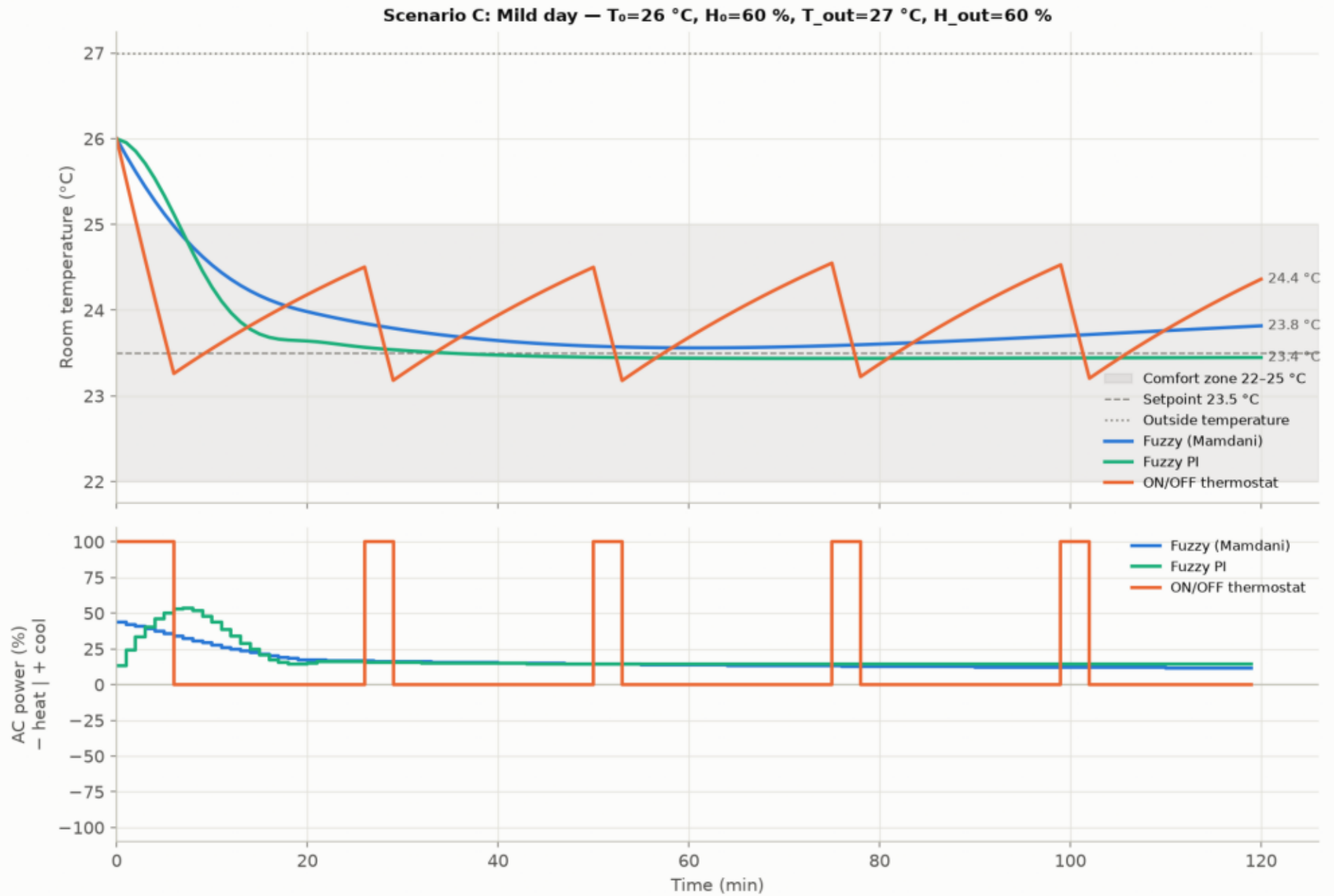
Simulation — scenario A



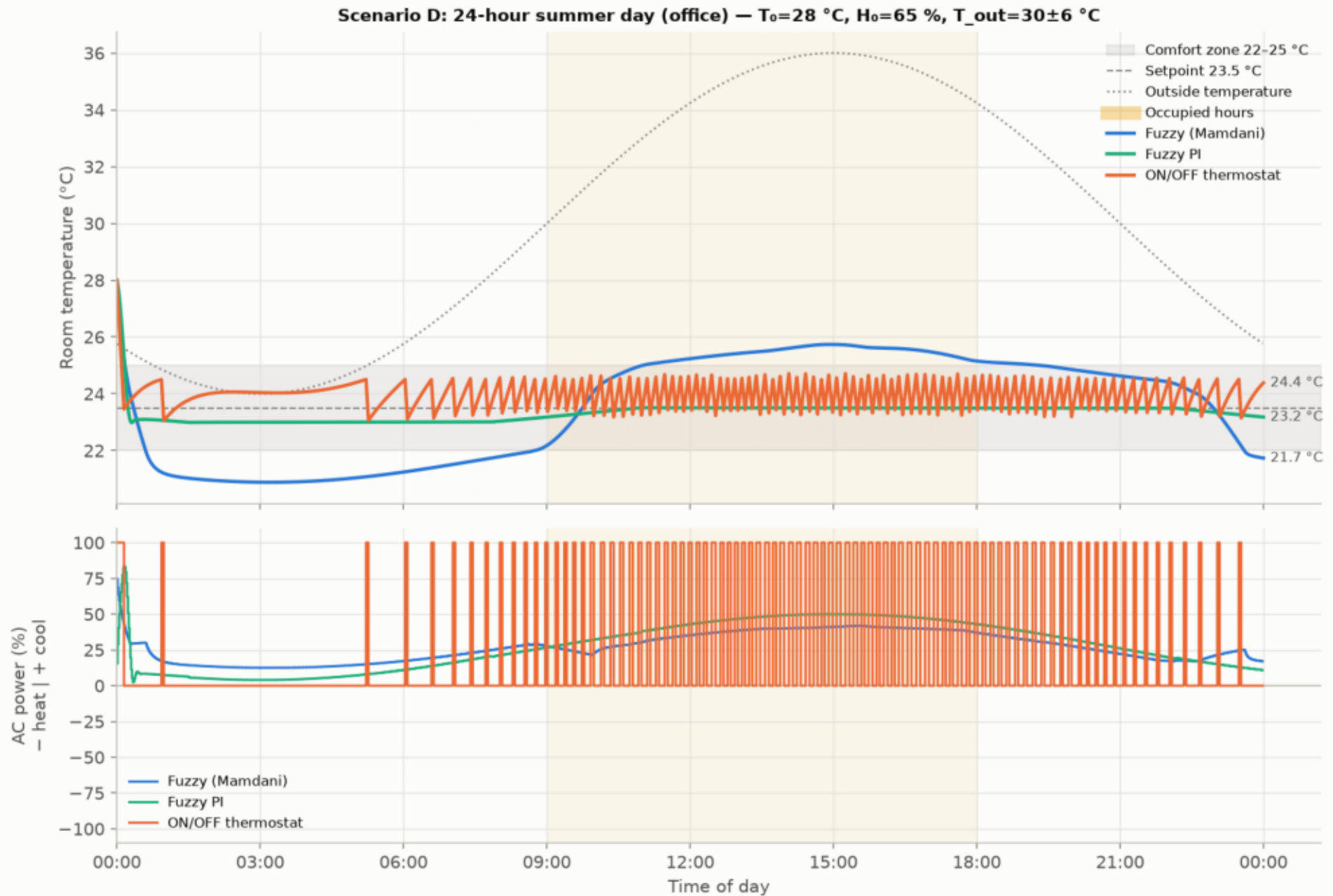
Simulation — scenario B



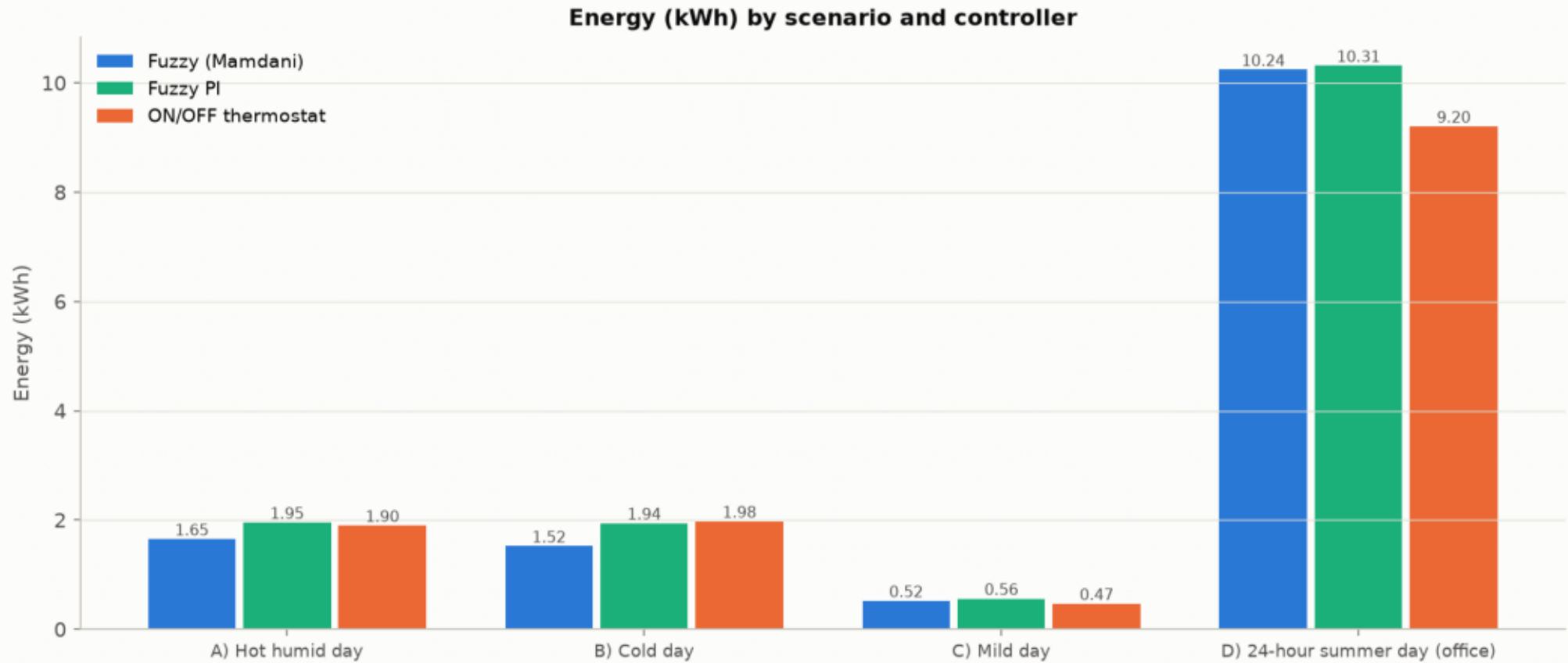
Simulation — scenario C



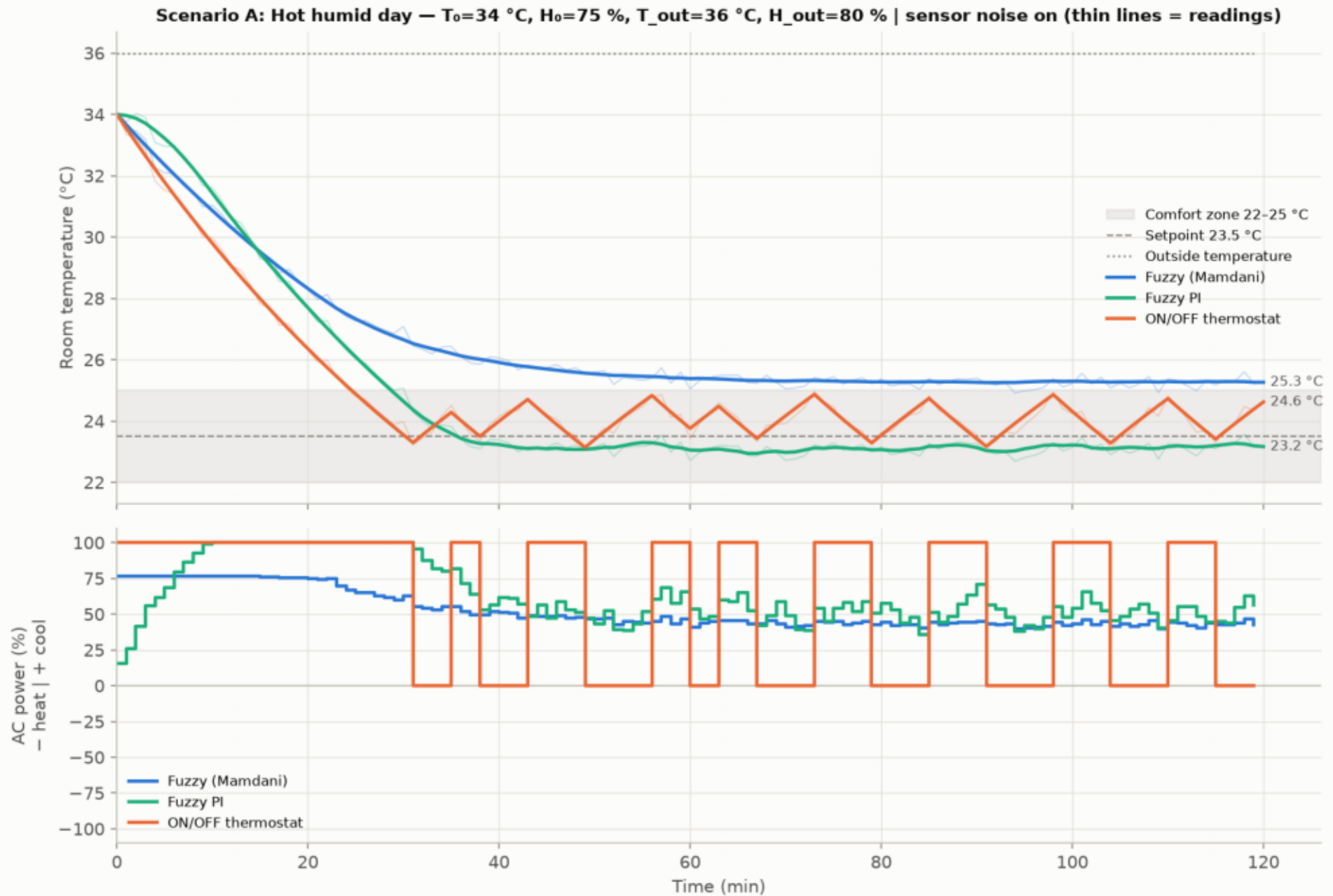
Simulation — scenario D



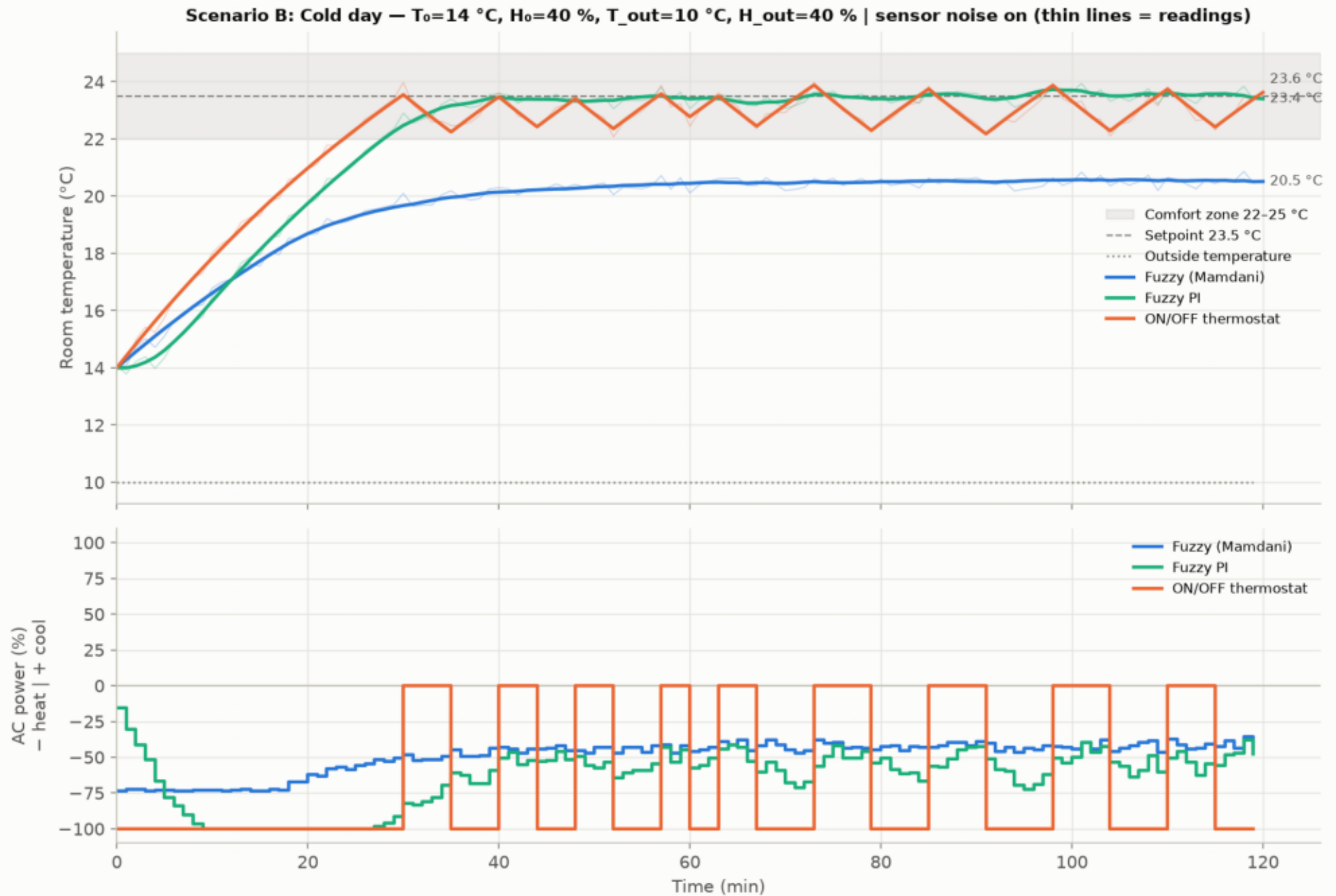
Electrical energy by scenario and controller



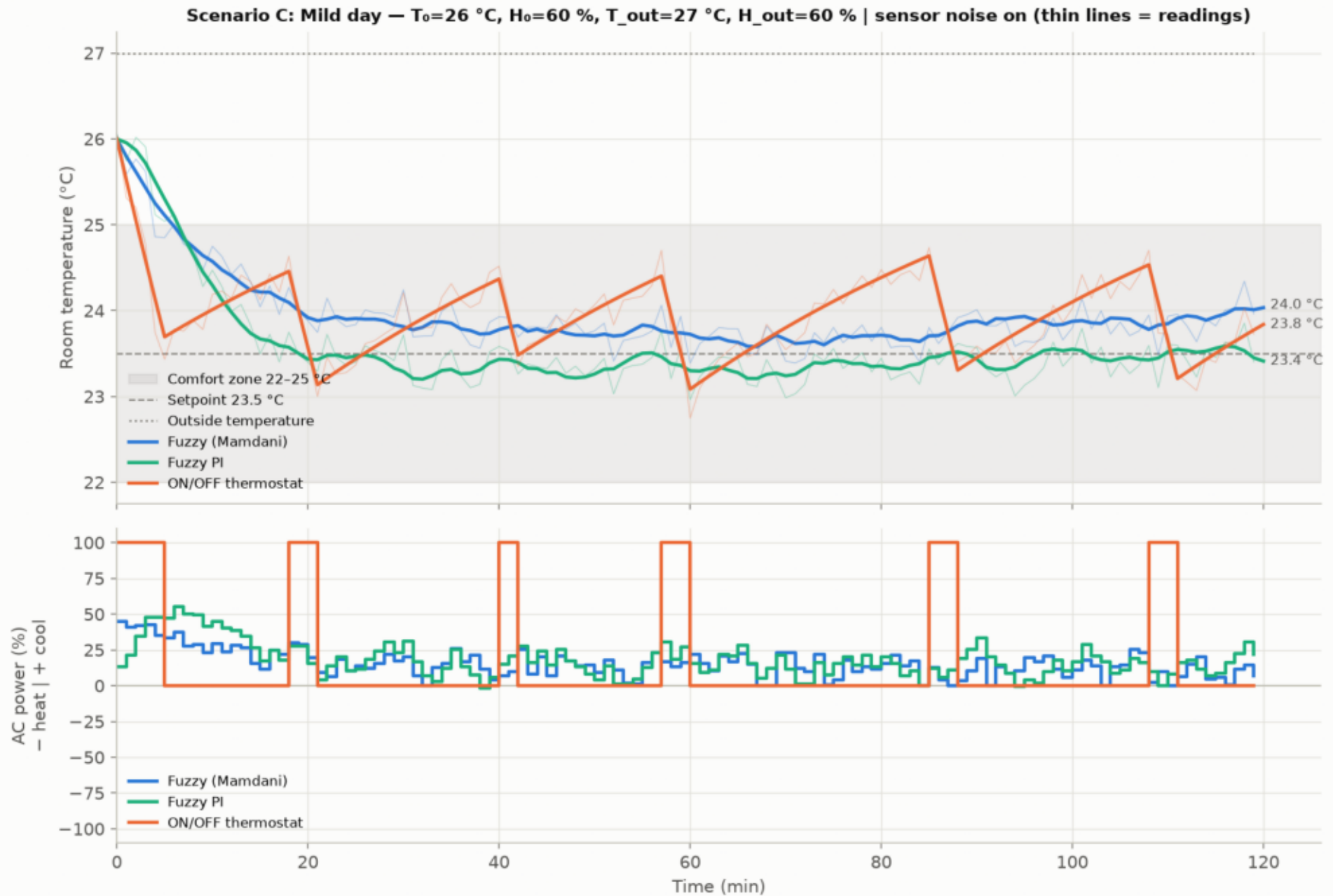
Simulation — scenario A noise



Simulation — scenario B noise



Simulation — scenario C noise



Simulation — scenario D occupancy

